RESEARCH PROPOSAL · NEURO-ONCOLOGY IMAGING AI

Automated Brain Tumor Segmentation in Multi-Class MRI Scans

Using a Hybrid CNN–Transformer U-Net (TransUNet) optimized with the Lion optimizer and Blue Whale Optimization (BWO)

Working title: A Comparative and Explainable Deep-Learning Framework for Brain Tumor Detection, Classification, and Segmentation in MRI

SECTION 01

Objectives

PRIMARY GOAL

Accurately segment brain tumors from contrast-enhanced T1-weighted MRI scans across three major pathological types — Glioma, Meningioma, and Pituitary Tumor.

CORE TECHNICAL CHALLENGES ADDRESSED

1

Local Detail vs. Global Context

Resolving the trade-off between fine anatomical boundaries (blurred by pure ViTs) and long-range semantic context (missed by local CNN receptive fields).

2

Training Stability & Memory

Accelerating and stabilizing attention-block weight updates during training while minimizing GPU memory footprint.

3

Class Imbalance

Overcoming extreme foreground–background imbalance through hybrid loss formulation and metaheuristic threshold optimization.

Dataset & Preprocessing Pipeline

Input data: Multi-sequence T1-contrast-enhanced brain MRI slices, categorized into Glioma, Meningioma, and Pituitary tumor classes.

01

Standardized Resizing

Spatially resize every scan to a uniform 224 × 224 × 3 input tensor.

02

Joint Augmentation

Synchronized horizontal/vertical flips and affine rotations applied identically to scan and mask.

03

Normalization & Masking

ImageNet feature standardization; binary mask encoding (1.0 = tumor, 0.0 = healthy tissue).

Network Architecture — Hybrid TransUNet

1

CNN Feature Extractor

Multi-stage convolutional backbone retains localized edge gradients, tumor margins, and texture — generating skip connections.

2

ViT Bottleneck

Reshapes the 56×56 spatial bottleneck into tokenized patches; Multi-Head Self-Attention captures global dependencies across distant tissue.

3

Cascaded U-Net Decoder

Upsamples context-rich transformer features and fuses them with high-resolution CNN skip connections to restore sharp boundaries.

Skip-connection fusion: high-resolution CNN features are concatenated at every decoder stage, preserving tumor-margin precision that pure transformer or pure CNN models lose independently.

Optimization Strategy

TRAINING · BACKPROPAGATION

Lion Optimizer

1. Replaces traditional AdamW for deep network parameter updates.
2. Uses sign operations with tracked momentum buffers — lower memory overhead and superior regularization on transformer attention layers.
3. Paired with a Cosine Annealing learning-rate schedule.

POST-PROCESSING · CALIBRATION

Blue Whale Optimization

1. Population-based swarm metaheuristic inspired by whale foraging — encircling prey and spiral bubble-net feeding.
2. Dynamically searches and converges on the optimal decision threshold $\tau \in [0.1, 0.9]$ on model output probabilities.
3. Maximizes validation Dice Similarity Coefficient (DSC), surpassing static 0.5 cutoffs on imbalanced tumor regions.

Objective Function

Hybrid Dice + Binary Cross-Entropy Loss

$$L_{total} = \lambda_{BCE} \cdot L_{BCE} + \lambda_{Dice} \cdot L_{Dice}$$

Pixel-Level Term

Log-loss penalization on every pixel prediction, driving precise per-pixel classification accuracy.

Region-Level Term

Overlap maximization between predicted and ground-truth tumor regions, directly optimizing the Dice metric.

Together, the two terms directly target the extreme foreground–background class imbalance inherent to tumor segmentation.

SECTION 03

Expected Outcomes

Dimension	Standard Pipeline (U-Net + AdamW)	Proposed Pipeline (TransUNet + Lion + BWO)
Global Context Modeling	Limited by fixed CNN receptive field	Full long-range modeling via ViT self-attention
Convergence Speed & VRAM	High VRAM — tracks 2 momentum states in AdamW	Lower VRAM footprint, faster convergence via Lion
Decision Thresholding	Fixed, arbitrary threshold (0.50)	Adaptively tuned to class distribution via BWO
Boundary Delineation	Susceptible to irregular margin errors	Refined via skip-connection feature fusion
Target Metrics	Baseline Dice: ~0.82–0.85	**Target Dice: > 0.90 · IoU: > 0.82**

SECTION 03 · TARGET PERFORMANCE

A More Accurate, Efficient, and Clinically Robust Framework

> 0.90

Target Dice Similarity Coefficient

vs. ~0.82–0.85 baseline

> 0.82

Target IoU (Jaccard Index)

region-overlap accuracy

↓ VRAM

Memory Footprint

via Lion optimizer

↑ Speed

Convergence Rate

faster, more stable training

Hybrid TransUNet unites CNN precision with Transformer context — calibrated end-to-end for reliable, clinically-meaningful tumor segmentation.